

CIT-421 Computer Graphics and Image Processing

Professor Md Abdul Masud

References

- Book-1: [Maria Petrou, Costas Petrou - Image Processing : The Fundamentals \(Second Edition\)](#)
- Book-2: [Chris Solomon, Toby Breckon - Fundamentals of Digital Image Processing : A Practical Approach with Examples in MATLAB](#)
- Book-3: [Rafael C. Gonzalez, Richard E. Woods - Digital Image Processing \(4th Edition\)](#)
- Book-4: [Wilhelm Burger, Mark J. Burge - Principles of Digital Image Processing : Fundamental Techniques](#)
- Book-5: Pixels : Basis of all Image Processing

Chapter-1 Representation (Book-2)

1.1 What is an Image?

- Image Representation
- Resolution
 - Spatial
 - Temporal
 - Bit
 - Binary → 2 bits
 - Grayscale → 8 bits
 - Color → 24 bits

1.2.1 Bit Plane Slicing

fig 1.3

1.2 Image Formats

1.3.1 Image Data Types

1.4 Color Space

RGB → Grayscale conversion

Hue-Saturation Value graph

1.5 Lab → Lab Work (MATLAB/Python)

1.5.1

Example 1.1

1.5.2

Example 1.2

1.5.3

Accessing pixel values

Example 1.5

Chapter-1 Introduction (Book-1)

Slide: [Lecture 01 - Introduction Image Processing](#)

- What is an image / digital image?
- How is a digital image formed?

Example 1.1

Fig 1.3

- How many bits do we need to store an image?

$2^m \rightarrow$ grayscale

$$2^m = 256, m = 8$$

$N \times N$ Resolution

$$b = N \times N \times m$$

- What determines the quality of an image?
 - What makes an images blurred?
 - What is meant by image resolution?
 - What does contrast mean?
 - What does good contrast mean?
 - What is the purpose of image processing?
 - How do we do image processing?
 - What is a linear operator?
 - How does a linear operator treat an image?
 - What is the meaning of point spread function?
-

Chapter 3 : Histogram (Book-4)

- To depict image statistics

3.1 What is a Histogram?

Histogram effects (topics in general)

Figure 3.3

3.2

3.2.1

- Exposure
- Contrast
- Dynamic Range

3.2.2 Image Effects

- Saturation
- Spikes and Gaps
- Impacts of Image compression
- Color Quantization

3.3 Computing Histogram

code implemetation

3.4

(self study)

3.5

(self study)

Chapter 3 : Pixels (Book-2)

Slide: [Lecture 02 - Pixel and Image Transformation](#)

Pixel transform, distribution of Pixels

3.1 What is Pixel?

Pixel is an abbreviation of "picture element" indexed an (x,y) column(c)-row(r)

- Spatial resolution
- Quantization level

Types of image - (study in details)

1. Color/Gray image
2. Infrared image (IR) → IR light reflection/ IR radiation
3. Medical Imaging - CT imaging, MRI imaging
4. Radar/Solar imaging
5. 3D imaging
6. Scientific imaging

3.2 Operation upon pixels

- Point transform
- Arithmetic operation
- Logical operation

One-one function mapping

I_A, I_B, C

$$I_{output} = I_A + I_B = I_A(i, j) + I_B(i, j) = I_A + C = I_A(i, j) + C$$

$$(i, j) = \{0, 1, 2, \dots, C - 1\} \{0, 1, 2, \dots, R - 1\}$$

Arithmetic Operation:

- Contrast adjustment
- Blending (self study and codeing)
- Subtraction, Multiplication, Division

! Assignement

Code implementation of arithmetic operations:

- Contrast adjustment
- Blending
- Subtraction
- Multiplication
- Division

3.2.2 Logical operation

NOT, XOR, OR, AND, Bitwise operation

NOT: Inversion $BU \rightarrow FG$

$$I_{output}(i, j) = MAX - I_{input}(i, j)$$

Image data type (previous class)

3.2.3 Thresholding (Slide)

fig 3.4 fig 3.5 (self study)

3.3 Point-based (slide)

- Functional transformation
- Dynamic range ★ ★ ★

3.3.1 Logarithmic transform (slide)

Exponential transformation (slide)

3.4 Pixel Distributions: Histograms

3.4.1 Histogram for Threshold Selection (Book + Slide)

fig 3.13

3.4.2 Adaptive Thresholding (slide)

3.4.3 Contrast Stretching (slide)

3.4.4 Hologram Equalization (slide)

! Assignment

- to code all from the book

Chapter 4 : Image Enhancement (Book-1)

Slide: [Lecture 03 - Image Enhancement](#)

What is image enhancement?

- noise
- gaussian noise
- impulse noise

Chapter 4 : Enhancement (Book-2)

4.2 Pixel neighbourhoods

4.3 Filter kernels and the mechanics of linear filtering

PPT/Slide : Linear Filtering

Example 4.2 ★ ★ ★

PPT/Slide : Non-linear Filtering

4.4 Filtering for noise removal

Figure 4.3

4.4.1 Mean filtering

Figure 4.4

4.4.2 Median filtering

Figure 4.5

4.4.3 Rank filtering

4.4.4 Gaussian filtering

4.5 Filtering for edge detection (Book-2 + PPT)

4.5.1 Derivative filters for discontinuities (Book-2)

Table 4.1

4.5.2 First-order edge detection (Book-2)

Figure 4.9

4.5.2.1 Linearly separable filtering

4.5.3 Second-order edge detection

Chapter 8 : Image Compression and Watermarking (Book-3)

Fundamentals

Equation 8.2

Three principal types of Data redundancy-

- Coding redundancy
- Spatial and temporal redundancy
- Irrelevant information

Coding Redundancy

Exercise 8.1

Spatial and Temporal Redundancy

Histogram (Fig 8.2)

Irrelevant Information

Fig 8.3

Measuring Image Information

Equation 8.5, 8.6, 8.7

Exercise 8.2

Shanon's First Theorum

Equation 8.9, 8.10, 8.11

Example 8.3

Image Compression Models (i)

Figure 8.5

Image Formats, Containers, and Compression Standards

8.2 Huffman Coding

Slide: [Huffman Coding](#)

Example

YouTube ()

Chapter 10 : Image Segmentation (Book-3)

Segmentation Algorithm- two basic properties of image intensity ...

10.1 Fundamentals

Section 2.5

Figure 10.1

10.2 Point, Line and Edge Detection

Table 10.1

(almost all equations)

Figure 10.2

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Reference Books

1. Zhigang Xiang, Roy A. Plaster - Schaum's Outline of Computer Graphics (2nd Edition)
2. Gonzales, Woods - Digital Image Processing (4th Edition)

Resource: [Tpoint Tech](#)

Mark Distribution

3 Sets

Slide: 19-33 Visual Computing (for idea only, irrelevant for exam)

Computer Graphics Tutorial

Computer Graphics

- [Computer Graphics Tutorial](#)
- [Interactive and Passive Graphics](#)

DPI → Dot Per Inch

Graphic System

- [Display Processor](#)
- [Cathode Ray Tube \(CRT\)](#)
- [Random Scan vs Raster Scan](#)
- [Color CRT Monitors](#)
- [Direct View Storage Tubes](#)
- [Flat Panel Display](#)

Scan Conversion a Line

- [Scan Conversion Definition](#)
- [Scan Converting a Point](#)
- [Scan Converting a Straight Line](#)
- [DDA Algorithm](#)
- [Bresenham's Line Algorithm](#)